EXPERIENTIAL LEARNING PROGRAM

EduSpend & Task Tracker (MERN Stack)

A Hybrid Academic Task & Personal Expense Management Web Application

1. Student & Submission Details

Student Name	**ABHISHEK KUMAR SINGH**
Enrollment Number	0157CS253D01
Course Track	Full Stack Development – MERN Stack Track
Training Partner	SmartBridge / FSP Experiential Learning
Submission Status	**Completed / Ready for Review**

2. Executive Summary

This project was developed as part of the **SmartBridge Experiential Learning Program** in Full Stack Web Development. The primary objective of *EduSpend* is to solve a common problem faced by engineering students: managing academic deadlines alongside daily personal expenses in a single unified dashboard. Built using the **MERN Stack** (MongoDB, Express.js, React.js, and Node.js), this application enables real-time record creation, retrieval, and management with a structured RESTful architecture.

3. Core Architectural Modules

Frontend (React.js)

Offers an interactive Single Page Application (SPA) user interface. Handles user input, state updates via React Hooks (`useState`, `useEffect`), and renders real-time data feeds.

Backend API (Node/Express)

RESTful server handling HTTP requests (GET, POST, DELETE). Implements CORS middleware and manages business logic for data operations.

Database (MongoDB)

NoSQL document store utilizing Mongoose schemas to safely store flexible task logs and expense records with automatic timestamping.

HTTP Communication (Axios)

Facilitates asynchronous API calls between client and server, handling JSON payloads for smooth CRUD operations.

4. Technical Implementation & Source Code

Below are the core source modules written during the Experiential Learning Program for the application's backend server and frontend user interface.

A. Express.js Server Implementation (`server.js`)

```
      const express = require('express'); const mongoose = require('mongoose'); const cors = require('cors'); const app = express(); app.use(express.json()); app.use(cors()); mongoose.connect('mongodb://127.0.0.1:27017/eduspend_db') .then(() => console.log("MongoDB Connected")) .catch(err => console.log(err)); const RecordSchema = new mongoose.Schema({ title: { type: String, required: true }, category: { type: String, required: true }, amount: { type: Number, default: 0 }, date: { type: Date, default: Date.now } }); const Record = mongoose.model('Record', RecordSchema); app.get('/api/records', async (req, res) => { const records = await Record.find().sort({ date: -1 }); res.json(records); }); app.post('/api/records', async (req, res) => { const newRecord = new Record(req.body); await newRecord.save(); res.json(newRecord); }); app.delete('/api/records/:id', async (req, res) => { await Record.findByIdAndDelete(req.params.id); res.json({ message: "Record deleted" }); }); app.listen(5000, () => console.log("Server running on port 5000"));
```

B. Key Learnings & Outcomes

- Gained practical experience with **Asynchronous JavaScript** and REST API integration.
- Understood full-stack state management using React functional components and lifecycle hooks.
- Configured Express middleware, MongoDB connectivity, and NoSQL schema modeling.
- Learned practical debugging and development workflows recommended during SmartBridge sessions.

5. Student Verification

I hereby confirm that this project report represents my original work completed under the SmartBridge MERN Stack track for the Experiential Learning assessment.

Date: October 30, 2026
Location: College Campus

ABHISHEK KUMAR SINGH
Signature of Student